WORKED SOLUTION

STEP I 2005, Question 4

Pure • Multiple Angle Formulae

STEP I 2005, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q4

The key idea

Both parts are about multiple angles. In (i) the only real decision is the sign of $\sin\theta$, which the given range settles; after that the double- and triple-angle formulae do the work. In (ii) we build $\tan 3\theta$ as $\tan(2\theta+\theta)$, and then the "hence" turns into a cubic in $t=\tan\theta$. A cubic in a STEP question nearly always has a root you can spot, and the range for θ tells us which root to keep.

Solution

Part (i)

Since $\cos^2\theta+\sin^2\theta=1$,

$$\sin^2\theta = 1-\tfrac{9}{25} = \tfrac{16}{25}, \qquad \text{so} \qquad \sin\theta = \pm\tfrac{4}{5}.$$

For $\frac{3\pi}{2}\le\theta\le 2\pi$, θ lies in the fourth quadrant, where $\sin\theta\le 0$. So $\sin\theta=-\frac{4}{5}$, and

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\left(-\tfrac{4}{5}\right)\left(\tfrac{3}{5}\right) = -\frac{24}{25},$$

as required.

For $\cos 3\theta$ we need the triple-angle formula. Writing $c=\cos\theta$ and $s=\sin\theta$,

$$\begin{aligned}\cos 3\theta=\cos(2\theta+\theta) &= \cos 2\theta\cos\theta-\sin 2\theta\sin\theta\\ &= (2c^2-1)c-2s^2c\\ &= (2c^2-1)c-2(1-c^2)c\\ &= 4c^3-3c.\end{aligned}$$

With $c=\frac{3}{5}$,

$$\cos 3\theta = 4\cdot\frac{27}{125}-\frac{9}{5} = \frac{108-225}{125} = -\frac{117}{125}.$$

(Here the sign of $\sin\theta$ doesn't matter, because $\cos 3\theta$ depends only on $\cos\theta$.)

Part (ii)

The identity. Write $t = \tan\theta$. By the double-angle formula, $\tan 2\theta = \dfrac{2t}{1-t^2}$, and by the addition formula,

$$\tan 3\theta = \tan(2\theta+\theta) = \frac{\tan 2\theta + \tan\theta}{1 - \tan 2\theta \tan\theta} = \frac{\dfrac{2t}{1-t^2} + t}{1 - \dfrac{2t^2}{1-t^2}}.$$

Multiplying the top and bottom by $1-t^2$,

$$\tan 3\theta = \frac{2t + t(1-t^2)}{(1-t^2) - 2t^2} = \frac{3t - t^3}{1 - 3t^2},$$

which is the identity. The derivation needs $\tan 2\theta$ to exist, so $t^2 \neq 1$; when $t = \pm 1$ we can check directly: for $\theta = \frac{\pi}{4}$, say, the right-hand side is $\frac{3-1}{1-3} = -1 = \tan\frac{3\pi}{4}$, and $t = -1$ is the same with signs flipped.

Solving $\tan 3\theta = \frac{11}{2}$. By the identity,

$$\frac{3t - t^3}{1 - 3t^2} = \frac{11}{2} \quad\Longrightarrow\quad 6t - 2t^3 = 11 - 33t^2 \quad\Longrightarrow\quad 2t^3 - 33t^2 - 6t + 11 = 0.$$

Multiplying out is safe because $1 - 3t^2 \neq 0$ here: if it were zero, $\tan 3\theta$ would be undefined, not $\frac{11}{2}$.

Ask yourself How do we find a root of $2t^3 - 33t^2 - 6t + 11$?

Any rational root p/q in lowest terms has p dividing 11 and q dividing 2, so we only need to try $\pm 1, \pm 11, \pm\frac{1}{2}, \pm\frac{11}{2}$. Trying $t = \frac{1}{2}$:

$$2\cdot\tfrac{1}{8} - 33\cdot\tfrac{1}{4} - 3 + 11 = \tfrac{1}{4} - \tfrac{33}{4} + 8 = 0.$$

So $2t - 1$ is a factor, and comparing coefficients gives

$$2t^3 - 33t^2 - 6t + 11 = (2t-1)(t^2 - 16t - 11).$$

The quadratic has roots $t = 8 \pm \sqrt{64+11} = 8 \pm 5\sqrt{3}$. So the three candidates are

$$t = \tfrac{1}{2}, \qquad t = 8 + 5\sqrt{3} \approx 16.66, \qquad t = 8 - 5\sqrt{3} \approx -0.66.$$

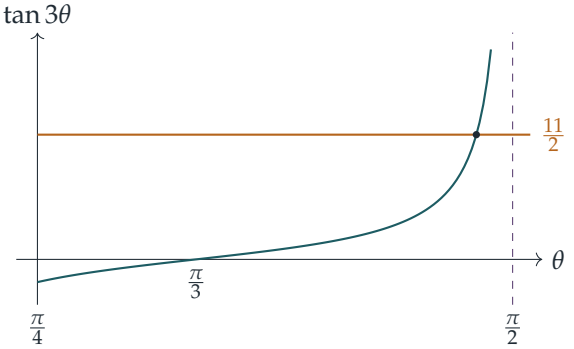
These correspond to the three angles in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ whose triple has tangent $\frac{11}{2}$.

Now use the range. Since tan is increasing on $\left[\frac{\pi}{4}, \frac{\pi}{2}\right)$ and $\tan\frac{\pi}{4} = 1$, we need $t \geq 1$ (and $\theta = \frac{\pi}{2}$ itself is excluded, as $\tan\theta$ is undefined there). Only one candidate qualifies:

$$\tan\theta = 8 + 5\sqrt{3}.$$

The other two are genuine roots of the cubic but belong to angles outside the range: $\frac{1}{2} < 1$ and $8 - 5\sqrt{3} < 0$.

The graph confirms there is exactly one solution: as θ runs from $\frac{\pi}{4}$ to $\frac{\pi}{2}$, 3θ runs from $\frac{3\pi}{4}$ to $\frac{3\pi}{2}$, and on that interval $\tan 3\theta$ increases continuously from -1 to $+\infty$, so it crosses $\frac{11}{2}$ exactly once.



Checks

- In (i), $\theta = 2\pi - \arccos\frac{3}{5} \approx 5.3559$, and a calculator gives $\sin 2\theta = -0.96$ and $\cos 3\theta = -0.936 = -\frac{117}{125}$.
- In (ii), $\theta = \arctan(8 + 5\sqrt{3}) \approx 1.5108$, which is between $\frac{\pi}{4} \approx 0.785$ and $\frac{\pi}{2} \approx 1.571$, and $\tan 3\theta \approx 5.5000$.
- A quick test of the identity: at $\theta = \frac{\pi}{12}$, $t = 2 - \sqrt{3}$, and the right-hand side simplifies to $1 = \tan\frac{\pi}{4}$.

Common slips

- Taking $\sin\theta = +\frac{4}{5}$ in (i), which gives the wrong sign for $\sin 2\theta$.
- Stopping at $t = \frac{1}{2}$ in (ii) because it is the root you spotted. It is a root of the cubic, but $\arctan\frac{1}{2} < \frac{\pi}{4}$.

Beyond STEP

The cubic had three roots because $\tan 3\theta = \frac{11}{2}$ has three solutions for θ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, spaced $\frac{\pi}{3}$ apart. The same mechanism runs in reverse in the classical trigonometric solution of cubics, and more generally de Moivre's theorem gives $\tan n\theta$ as a ratio of polynomials in $\tan\theta$ whose roots are the tangents of equally spaced angles. Needham's *Visual Complex Analysis* shows why multiple-angle formulae come out of $(\cos\theta + \mathrm{i}\sin\theta)^n$ so naturally.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.